ONE

PRISTINE WATER

BY PERKUP

ONE

Scientific White Paper

The Science Behind ONE Technology

INTRODUCTION

Water has always been regarded as the most essential element for sustaining life. However, over the past few decades, discussions about water quality have focused almost exclusively on the removal of contaminants, while giving far less attention to an equally important aspect: **its natural mineral composition**.

Much of the water currently available—whether sourced from underground aquifers, public distribution systems, or purification processes—has a mineral profile that differs significantly from that found in nature in its most pristine state. In many cases, naturally occurring minerals are removed during treatment and later reintroduced artificially, primarily to meet physicochemical standards or improve taste.

ONE Technology was developed based on a different philosophy.

Rather than simply desalinating seawater, our objective has been to understand how to preserve the naturally occurring mineral balance of the oceans while removing excess salts and potential contaminants, maintaining the greatest possible fidelity to the extraordinarily complex mineral system that nature has spent billions of years creating.

The ocean is far more than the largest body of water on Earth.

It is the planet's greatest natural chemical laboratory.

Over billions of years, virtually every naturally occurring element found within the Earth's crust has undergone continuous processes of erosion, dissolution, river transport, volcanic activity, sedimentation, precipitation, and geochemical recycling before ultimately reaching the oceans. This perpetual cycle transformed seawater into an exceptionally complex and stable system containing macroelements, microelements, and trace elements distributed within a natural equilibrium that is extraordinarily difficult to reproduce artificially.

This characteristic fundamentally distinguishes water produced through the simple addition of mineral salts from water whose composition derives from a natural system shaped throughout Earth's geological history.

It is important to understand that nature does not operate through isolated elements.

Just as the human body depends upon the simultaneous interaction of hundreds of biochemical processes, minerals also function as an integrated system. Modern science demonstrates that concentration, chemical form (speciation), bioavailability, and interactions among different elements directly influence the behavior of every substance within biological systems.

For this reason, evaluating any mineral in isolation may lead to incomplete conclusions. The scientific assessment of water must consider its entire chemical composition as an integrated system rather than focusing solely on the presence or absence of specific elements.

Another fundamental principle that guided the development of ONE Technology is one of the cornerstones of modern toxicology, established more than five centuries ago by Paracelsus:

"All things are poison, and nothing is without poison; only the dose makes a thing not a poison."

This principle remains fully valid in contemporary science. The safety of any chemical element depends upon factors such as its concentration, chemical form, exposure time, bioavailability, and interaction with other components present in the environment.

Throughout this document, we present the scientific principles that guided the development of ONE Water, together with responses to the most common technical questions regarding seawater-derived drinking water.

Our objective is not to claim that water alone replaces a balanced diet or medical treatment, nor to attribute therapeutic properties without scientific evidence.

Rather, the purpose of this document is to demonstrate that nature has developed, over billions of years, an extraordinarily complex and balanced mineral system, and that ONE Technology was created to preserve this natural richness while maintaining the highest standards of safety, purity, and quality.

More than simply producing desalinated water, our mission is to preserve one of the most complete natural mineral compositions found on Earth, offering water that respects the intelligence of nature itself.

2. WHY THE OCEAN IS THE WORLD'S GREATEST NATURAL CHEMICAL LABORATORY

To understand the philosophy behind ONE Technology, it is first necessary to understand the role of the oceans throughout Earth's history.

Long before the emergence of humanity—and even before the first complex forms of life—the oceans already functioned as a vast natural system of chemical transformation and equilibrium. For approximately four billion years, the Earth underwent intense geological processes that shaped the chemical composition of seawater.

Rainfall, rivers, winds, volcanic activity, mountain erosion, the dissolution of minerals from rocks, tectonic movements, and biological activity have continuously transported chemical elements into the oceans. At the same time, countless natural processes have promoted precipitation, sedimentation, recycling, and redistribution of these elements, establishing a dynamic equilibrium that continues evolving to this day.

In other words, the oceans represent an immense geochemical system operating continuously.

No laboratory in the world could artificially reproduce billions of years of interactions among minerals, living organisms, ocean currents, the atmosphere, and geological processes.

It is precisely this extraordinary complexity that makes seawater unique.

Unlike solutions produced simply by dissolving mineral salts into purified water, seawater contains an exceptionally complex combination of macroelements, microelements, and trace elements coexisting within a natural balance developed throughout Earth's geological history.

This balance is not static.

It results from the continuous interaction of physical, chemical, and biological processes operating simultaneously throughout the oceans. Consequently, the mineral composition of seawater represents one of the most sophisticated natural chemical systems known to science.

It is important to recognize that nature rarely works through isolated elements.

Natural systems are built upon networks of interactions. One mineral influences the behavior of another; certain ions promote or inhibit the solubility of different compounds; numerous elements participate together in chemical and biological reactions that sustain the balance of marine ecosystems.

The human body itself operates according to this same principle. Human physiology depends upon the coordinated action of dozens of minerals and electrolytes working together to maintain essential processes such as nerve transmission, muscle contraction, osmotic balance, cellular metabolism, and countless enzymatic reactions.

For this reason, evaluating water quality should never be limited to the isolated presence of a specific chemical element. Scientific understanding requires observing the system as a whole, taking into account concentrations, chemical forms, interactions among minerals, and the context in which those elements exist.

This systemic perspective served as the foundation for the development of ONE Technology.

Rather than viewing seawater merely as a source of salt to be removed, our approach has been to preserve, as faithfully as possible, the naturally occurring mineral balance of the oceans while eliminating excess salts and potential contaminants introduced by human activity, without compromising the mineral richness created by nature itself.

In essence, ONE Technology is built upon a simple principle:

Nature has spent billions of years creating one of the most complex and balanced mineral systems on Earth. Our mission is not to reinvent it, but to preserve it with safety, purity, and scientific responsibility.

3. THE NATURAL MINERAL BALANCE OF THE OCEANS

When discussing minerals present in water, it is common to assume that the greater the number of minerals, the healthier the water must be.

However, this is an oversimplified view of an extraordinarily complex natural system.

In nature, what truly matters is not merely the presence of minerals, but the way in which they coexist in balance.

The oceans provide the finest example of this principle.

Over billions of years, geological, physical, chemical, and biological processes have established a relatively stable mineral composition in which thousands of interactions continuously occur among macroelements, microelements, and trace elements. This balance is dynamic, constantly adjusted by the Earth's natural cycles.

Within a natural system, no mineral acts entirely in isolation.

Each element influences, to varying degrees, the behavior of others. Some compete for the same absorption mechanisms, others promote specific chemical reactions, while some stabilize compounds and others participate in enzymatic processes essential to living organisms.

This continuous interaction makes the entire system far more important than the isolated analysis of any single element.

For this reason, modern science increasingly adopts a systemic approach when studying natural environments. Evaluating only the presence or absence of a particular mineral may lead to incomplete conclusions if the surrounding chemical context is ignored.

Furthermore, the same chemical element may behave in completely different ways depending upon its chemical form, concentration, bioavailability, and interactions with other components within the solution.

This principle is widely recognized in chemistry, biochemistry, and toxicology.

Just as an orchestra is defined not by a single instrument but by the harmony among all of them, the mineral composition of water should be understood as an integrated system in which every component contributes to a much broader natural equilibrium.

Nature itself offers countless examples of this principle.

Many plants contain compounds that, when isolated or present in inappropriate concentrations, may be harmful to the human body. Yet these same plants have given rise to medicines used in modern healthcare because their effects depend upon dosage, chemical form, methods of preparation, and the interaction among their naturally occurring compounds.

This same principle demonstrates why the scientific evaluation of water cannot be reduced to the simple identification of individual chemical elements.

Modern toxicology recognizes that no element should be classified as inherently "good" or "bad" without considering its concentration, chemical form, bioavailability, exposure time, and the environmental context in which it exists.

It was precisely this scientific perspective that guided the development of ONE Technology.

Our objective has never been simply to produce water containing a greater number of minerals.

Nor has it been to artificially add minerals after desalination.

Instead, our goal has been to preserve, as faithfully as possible, the naturally occurring mineral balance of seawater while removing excess salts and potential external contaminants, respecting the chemical architecture that nature itself has built over billions of years.

This concept represents a fundamental distinction from artificially remineralized waters.

Whereas conventional remineralization generally consists of adding a limited number of selected mineral salts after purification, ONE Technology seeks to preserve a naturally complex mineral composition whose organization results from geological and oceanographic processes that cannot be fully reproduced in a laboratory.

In essence, we believe that nature has already carried out the most sophisticated mineral balancing process known to science.

ONE Technology does not seek to replace that natural intelligence, but rather to preserve this extraordinary geochemical heritage with scientific rigor, safety, and respect for the limits established by science itself.

"The true richness of ocean water lies not in the number of minerals it contains, but in the extraordinary chemical harmony that nature has developed over billions of years. Preserving that balance—with safety and scientific responsibility—is the very essence of ONE Technology."

4. MACROELEMENTS, MICROELEMENTS, AND TRACE ELEMENTS

The mineral composition of seawater is the result of an extraordinarily complex natural system composed of elements present at different concentrations. Although the periodic table contains dozens of elements naturally found in the oceans, they are not distributed uniformly.

From a chemical perspective, these elements can be classified into three major groups: **macroelements, microelements, and trace elements**.

This classification does not indicate the biological importance of each element, but rather their relative concentration within the water.

4.1 Macroelements

Macroelements are those present in the highest concentrations and account for virtually the entire mineral composition of seawater.

The principal macroelements include:

- Chloride (Cl^-)
- Sodium (Na^+)
- Sulfate (SO_4^{2-})
- Magnesium (Mg^{2+})

- Calcium (Ca^{2+})
- Potassium (K^{+})

Together, these six components represent approximately **99% of all dissolved salts naturally present in seawater.**

Beyond their abundance, these elements play fundamental roles in maintaining the chemical balance of seawater and participate in countless biological processes throughout marine ecosystems.

Following the ONE desalination process, excess salts are removed to levels appropriate for human consumption while preserving the concept of mineral balance that characterizes the natural composition of ocean water.

4.2 Microelements

Microelements are present at significantly lower concentrations than macroelements, yet they remain an integral part of the ocean's natural mineral composition.

These may include elements such as:

- Iron
- Zinc
- Manganese
- Copper
- Molybdenum
- Strontium
- Boron
- Silicon

These elements participate in numerous geochemical and biological processes within marine ecosystems.

Their presence results from billions of years of interaction among rocks, rivers, volcanic activity, the atmosphere, and living organisms, naturally contributing to the ocean's chemical system.

4.3 Trace Elements

Trace elements are present in extremely low concentrations, typically measured in micrograms (µg/L) or nanograms (ng/L) per liter.

Their low concentration does not imply scientific insignificance.

On the contrary, many trace elements are widely used as environmental, geochemical, and oceanographic indicators, helping researchers better understand the natural processes occurring within marine environments.

It is important to emphasize that the mere analytical detection of an element does not automatically indicate a health risk.

Such an assessment depends upon several factors, including:

- The actual concentration present;
- Its chemical form (speciation);
- Bioavailability;
- Duration of exposure;
- Quantity consumed;
- Safety limits established by health authorities.

For this reason, interpreting a mineral analysis always requires an integrated evaluation of the complete chemical profile rather than simply identifying which elements are present.

4.4 An Integrated Chemical System

Although dividing minerals into macroelements, microelements, and trace elements is useful for educational purposes, these categories should not be viewed as independent systems.

In nature, they coexist within a single, integrated chemical equilibrium.

Macroelements influence the ionic strength of the solution.

Microelements participate in countless chemical reactions.

Trace elements contribute to the overall complexity of the natural system.

The result is a network of physicochemical interactions that has continuously evolved over billions of years and remains in dynamic equilibrium today.

This characteristic fundamentally distinguishes seawater from artificial solutions produced simply by adding selected mineral salts to purified water.

4.5 The Philosophy Behind ONE Technology

The development of ONE Technology began with the understanding that the richness of seawater lies not merely in the presence of specific minerals, but in the natural organization of the entire mineral system.

For this reason, our objective has never been simply to remove salt from seawater.

Instead, within the safety standards established for human consumption and in compliance with applicable quality regulations, our goal has been to preserve as much of the ocean's naturally occurring mineral complexity as possible.

More than a desalination process, this approach is founded upon preserving the mineral balance that nature has developed throughout Earth's geological history.

How Nature Organizes Its Minerals

Category	Relative Concentration	Examples	Role Within the System
Macroelements	High	Sodium, Chloride, Magnesium, Calcium, Potassium	Form the primary chemical structure of seawater
Microelements	Moderate	Iron, Zinc, Boron, Strontium	Participate in geochemical and biological processes
Trace Elements	Very Low	Various trace elements	Contribute to the natural complexity of the system

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5. MODERN TOXICOLOGY: WHY THE DOSE MAKES THE POISON

Many concerns regarding the mineral composition of water arise simply from the identification of specific chemical elements in laboratory analyses.

It is common to find interpretations that classify an element as "toxic" solely because it is present, without considering the fundamental principles of modern toxicology.

Although understandable, this approach does not reflect contemporary scientific knowledge.

Modern toxicology is based on a principle established nearly five centuries ago by the Swiss physician and alchemist **Paracelsus (1493–1541)**, widely regarded as the father of toxicology:

"All things are poison, and nothing is without poison; only the dose makes a thing not a poison."

This principle remains one of the cornerstones of toxicological risk assessment used today in medicine, pharmacology, chemistry, environmental science, and by regulatory agencies around the world.

In other words, no chemical element should be considered inherently safe or inherently hazardous simply because it is present. Its biological significance depends upon multiple factors that determine how it behaves within living systems.

5.1 Concentration Matters

The presence of a chemical element does not, by itself, imply a health risk.

The most important factor is the **actual concentration** at which that element is present.

Many elements that are essential to life may produce adverse effects when consumed in excessive amounts.

Likewise, elements that require toxicological attention may occur at concentrations far below internationally accepted safety limits.

For this reason, modern laboratory analyses do not merely identify which elements are present—they also determine their concentrations and compare them against internationally recognized drinking water standards.

5.2 Chemical Form (Speciation)

Another frequently overlooked aspect is that the same chemical element may exhibit completely different biological behaviors depending on the chemical form in which it exists.

This concept is known as **chemical speciation**.

The toxicity of an element depends not only on the element itself but also on the molecule or compound of which it forms part.

Consequently, two compounds containing the same element may exhibit entirely different biological properties.

For this reason, modern science does not evaluate elements in isolation but considers the chemical environment in which they occur.

5.3 Bioavailability

Not every element present in water is necessarily absorbed by the human body.

Its absorption depends upon several factors, including:

- Chemical form;
- Solubility;
- Interactions with other minerals;
- Individual physiological characteristics;
- Metabolism;
- Dietary conditions.

This concept is known as **bioavailability** and represents one of the fundamental principles of mineral nutrition and toxicology.

Therefore, the mere presence of a specific element does not necessarily mean that it will be absorbed in the same proportion by the human body.

5.4 Duration of Exposure

Another essential factor is the duration of exposure.

Modern toxicology distinguishes among:

- **Acute exposure** (short-term);
- **Subchronic exposure**;
- **Chronic exposure** (long-term).

This distinction is critical because toxicological risk depends not only on concentration but also on the frequency and duration of exposure over time.

For this reason, regulatory agencies establish safety limits based upon typical long-term patterns of human consumption.

5.5 The System Is More Important Than the Individual Element

Nature operates through complex systems.

No element exists in isolation within seawater.

Every element forms part of an extensive network of physicochemical interactions that has evolved over billions of years.

For this reason, the scientific evaluation of water quality must consider the system as a whole, taking into account concentrations, chemical equilibrium, mineral interactions, and compliance with drinking water quality standards.

Reducing this assessment to the simple identification of individual chemical elements represents an oversimplification that does not reflect how modern toxicology evaluates environmental and biological systems.

5.6 ONE Technology and Water Safety

ONE Technology was developed in accordance with these same scientific principles.

The desalination process is designed not only to remove excess salts but also to reduce or eliminate contaminants that could represent potential health risks, while maintaining rigorous analytical controls to ensure compliance with applicable drinking water quality standards.

Furthermore, every production batch may undergo laboratory analyses to verify its physicochemical composition and confirm that all safety parameters remain within the limits established by technical and regulatory standards.

Our commitment extends beyond preserving the ocean's natural mineral balance.

It is equally committed to doing so with scientific responsibility, transparency, and full respect for internationally recognized criteria governing water intended for human consumption.

Conclusion

Modern toxicology teaches that the mere presence of a chemical element is not sufficient to determine whether a substance represents a benefit or a risk.

Scientific evaluation requires the simultaneous consideration of concentration, chemical form, bioavailability, duration of exposure, and the environmental context in which the element exists.

These are the same principles that guide both contemporary science and the development of ONE Technology.

Preserving the ocean's natural mineral balance does not mean disregarding toxicology.

On the contrary, it means understanding toxicology in its entirety and applying its principles responsibly to produce water that combines mineral richness, purity, and safety within internationally recognized scientific and regulatory standards.

"If the natural presence of certain elements alone were sufficient to make the oceans incompatible with life, they could hardly have sustained the largest and most diverse ecosystem on Earth for billions of years. Science demonstrates that the safety of any natural system depends upon the balance of its composition, the concentrations

involved, and the absence of external contamination."

6. BIOAVAILABILITY: MORE IMPORTANT THAN THE QUANTITY OF MINERALS

When evaluating the composition of water, attention is often focused solely on the quantity of minerals it contains.

However, nutritional science has demonstrated that the mere presence of a mineral does not necessarily mean it will be effectively utilized by the human body.

For any nutrient to perform a biological function, it must first be available for absorption.

This concept is known as **bioavailability**.

In other words, it is not enough for a mineral simply to be present; it must also be capable of being absorbed, transported, and utilized by the body's cells.

This capacity depends upon several factors, including:

- Its chemical form;
- Its solubility;
- Interactions with other minerals;
- The pH of the surrounding environment;
- The individual's nutritional status;
- Individual metabolic characteristics.

For this reason, two waters containing similar concentrations of a particular mineral may exhibit different biological behavior.

Nature Operates Through Balance

Within natural systems, minerals do not exist in isolation.

They coexist within a highly complex chemical environment in which countless interactions occur simultaneously.

Some elements facilitate the absorption of others.

Others compete for the same transport mechanisms within the body.

Certain minerals contribute to the chemical stability of the solution, while others influence the availability of different nutrients.

These relationships are a natural part of chemistry and demonstrate that the composition of water should always be evaluated as an integrated system rather than as a collection of isolated components.

Water and Nutrition

It is important to emphasize that the primary function of water is to maintain proper hydration.

A balanced diet remains the principal source of vitamins, proteins, carbohydrates, fats, and minerals required by the human body.

ONE Technology is not based on the premise that water replaces nutrition or that, by itself, it provides all of the body's nutritional requirements.

Nevertheless, water serves as the medium through which countless physiological processes occur.

It is involved in nutrient transport, body temperature regulation, metabolic reactions, waste elimination, and the proper functioning of every cell.

When naturally present, dissolved minerals also become part of this physiological environment.

The Philosophy Behind ONE Technology

The objective of ONE Technology is not to artificially enrich water by adding mineral salts after desalination.

Our philosophy is fundamentally different.

We seek to preserve, within internationally recognized safety standards for human consumption, the ocean's naturally occurring mineral composition while respecting the balance that nature has developed over billions of years through Earth's geological processes.

This approach recognizes that the quality of water depends not only on the quantity of minerals present, but also on the way those minerals coexist within a naturally balanced system.

Science, Nature, and Responsibility

Bioavailability remains an active field of scientific research.

Although many aspects of mineral interactions continue to be investigated, there is broad scientific consensus that factors such as concentration, chemical form, solubility, and interactions among elements directly influence how nutrients are utilized by the human body.

It is upon this scientific foundation that ONE Technology has been developed.

Our commitment is to preserve the ocean's natural mineral richness responsibly, while respecting applicable quality and safety standards.

We do not attribute therapeutic properties to water.

Rather, we recognize the value of a mineral system that has been shaped by nature itself over the course of Earth's geological history.

Closing Statement

"Water is far more than a simple source of hydration. It is the medium through which life operates. When its natural mineral balance is preserved with scientific responsibility, it reflects one of nature's most extraordinary achievements."

7. MYTHS AND FACTS ABOUT SEAWATER AND ONE TECHNOLOGY

Throughout the development of ONE Technology, numerous questions have been raised regarding the mineral composition of seawater-derived drinking water.

These questions are both legitimate and important, reflecting the natural curiosity surrounding an innovative technology.

In this chapter, we address the most frequently asked questions using principles drawn from chemistry, toxicology, physiology, oceanography, and water science, always within the boundaries of current scientific knowledge.

Myth 1 — "If water contains more minerals, it must be healthier."

Clarification

Not necessarily.

Water quality cannot be determined solely by the quantity of minerals it contains.

What truly matters is the balance among those minerals, their concentrations, their chemical forms, and the absence of contaminants.

The philosophy behind ONE Technology has never been to produce water containing *more* minerals than other waters.

Our objective is to preserve, within internationally recognized safety standards for human consumption, a naturally balanced mineral composition shaped by the geological processes that have influenced the oceans for billions of years.

The true distinction lies not in the quantity of minerals, but in the origin and natural balance of the mineral system.

Myth 2 — "The presence of heavy metals means the water is toxic."

Clarification

This is one of the most common—and also one of the most oversimplified—interpretations.

Modern toxicology does not conclude that a health risk exists simply because a particular chemical element has been detected.

A proper scientific evaluation must simultaneously consider:

- The concentration present;
- The chemical form (speciation);

- Bioavailability;
- Duration of exposure;
- Regulatory safety limits established by health authorities.

It is also important to distinguish between two concepts that are often confused.

Naturally occurring elements are part of the ocean's geochemical composition.

Environmental contaminants, on the other hand, are introduced through human activities such as industrial discharge, mining, sewage, agricultural runoff, and other forms of pollution.

ONE Technology was specifically designed to preserve the ocean's naturally occurring mineral richness while employing advanced purification processes to remove contaminants and maintain rigorous quality control throughout production.

Myth 3 — "Minerals may overload the kidneys."

Clarification

The kidneys naturally regulate the body's water and electrolyte balance.

When drinking water complies with internationally recognized quality standards, there is no scientific evidence that its normal consumption alone causes kidney overload in healthy individuals.

Furthermore, many of the naturally occurring elements found in seawater exist only in extremely low concentrations, commonly referred to as **trace elements**.

Water safety is determined not simply by the presence of these elements, but primarily by their concentrations within the finished product.

For this reason, ONE Technology employs comprehensive laboratory analyses to ensure that all production batches comply with applicable drinking water quality standards.

Myth 4 — "Water only hydrates; minerals come exclusively from food."

Clarification

Water's primary role is indeed hydration.

A balanced diet remains the principal source of vitamins, proteins, fats, carbohydrates, and essential minerals.

However, this does not mean that naturally dissolved minerals in water are irrelevant.

Water participates in virtually every physiological process within the human body, serving as the medium for nutrient transport, metabolic reactions, electrolyte balance, temperature regulation, and waste elimination.

When naturally present and within safe concentrations, dissolved minerals also become part of this physiological environment.

Hydration and mineral composition are therefore complementary—not competing—concepts.

Myth 5 — "Artificially remineralized water is equivalent to ONE Water."

Clarification

Artificial remineralization generally consists of adding selected mineral salts after purification.

This is a well-established and widely accepted practice.

ONE Technology, however, follows a fundamentally different philosophy.

Rather than reconstructing a mineral profile artificially, we seek to preserve, within internationally recognized drinking water safety standards, the naturally occurring mineral balance of the ocean.

That balance is the product of billions of years of geological, oceanographic, and biological processes, resulting in a system far more complex than the simple combination of a limited number of mineral salts.

We do not claim that one approach is universally superior to another.

The distinction of ONE Technology lies in preserving, as faithfully as possible, a naturally occurring mineral composition while respecting its intrinsic complexity.

Myth 6 — "If potentially toxic elements exist in the ocean, all marine life should be contaminated."

Clarification

This conclusion oversimplifies the functioning of marine ecosystems.

The oceans have supported the largest and most diverse ecosystem on Earth for billions of years.

Millions of species have evolved within this naturally balanced chemical environment, demonstrating that the mere presence of certain elements does not automatically render an ecosystem toxic.

At the same time, science recognizes that some marine organisms may accumulate contaminants when exposed to polluted environments or because of their position within the food chain.

For this reason, it is essential to distinguish between the **natural chemical composition of seawater** and **pollution introduced by human activity**.

ONE Technology is based on precisely this distinction, employing purification processes designed to remove contaminants while continuously monitoring the quality and safety of the finished water.

Myth 7 — "A laboratory analysis alone determines whether water is healthy."

Clarification

Laboratory analysis is an essential scientific tool, but interpreting the results requires technical expertise.

A simple list of detected elements is not sufficient, by itself, to determine whether water is suitable for human consumption.

A complete evaluation must consider:

- Element concentrations;
- Regulatory limits;

- Chemical speciation;
- Microbiological quality;
- Potential contaminants;
- Total Dissolved Solids (TDS);
- pH;
- Chemical stability;
- Compliance with drinking water regulations.

Only an integrated evaluation of all these parameters allows a scientifically sound assessment of water quality.

For this reason, ONE Technology is founded upon continuous laboratory testing, strict quality control procedures, and compliance with internationally recognized technical standards.

Conclusion

Throughout this document, we have sought to demonstrate that the scientific evaluation of water extends far beyond simply counting minerals or identifying individual chemical elements.

Water is a complex natural system whose quality depends upon the balance of its composition, the absence of contaminants, compliance with health standards, and an integrated understanding of chemistry, toxicology, physiology, and oceanography.

ONE Technology was developed based upon this systemic scientific perspective.

Our commitment is to preserve the ocean's natural mineral richness responsibly, removing excess salts and contaminants while respecting both the intelligence of nature and the highest internationally recognized standards for drinking water quality.

8. SCIENTIFIC FOUNDATIONS OF ONE TECHNOLOGY

ONE Technology was developed upon scientific principles that are widely recognized in the fields of oceanography, geochemistry, environmental chemistry, toxicology, human physiology, and water treatment technology.

Although the technology itself is the result of proprietary research and development, the scientific concepts that support its design are firmly grounded in well-established knowledge accumulated through decades of research across these disciplines.

The following principles form the scientific foundation of ONE Technology.

8.1 The Chemical Composition of the Oceans Is the Result of Billions of Years of Geological Evolution

Modern science recognizes that the mineral composition of seawater is the product of natural processes that have operated continuously since the formation of the Earth.

Rock weathering, volcanic activity, river transport, atmospheric interactions, the hydrological cycle, and biological processes have continuously transported and redistributed chemical elements over billions of years, giving rise to one of the most complex geochemical systems on the planet.

This understanding represents one of the fundamental principles of modern oceanography.

8.2 Toxicity Depends on Dose and Chemical Context

Contemporary toxicology recognizes that no substance should be considered inherently beneficial or inherently harmful simply because it is present.

Scientific evaluation requires consideration of factors such as:

- Concentration;
- Chemical form (speciation);
- Bioavailability;
- Duration of exposure;
- Frequency of consumption;
- Interactions with other elements.

Originally established by Paracelsus during the sixteenth century, this principle continues to serve as one of the foundations of modern toxicological science and regulatory risk assessment worldwide.

8.3 Water Is an Integrated Chemical System

Modern chemistry demonstrates that natural aqueous solutions should not be interpreted simply as collections of isolated chemical elements.

Minerals dissolved in water interact continuously, influencing chemical equilibrium, solubility, ionic strength, stability, and numerous other physicochemical properties.

For this reason, the scientific evaluation of water quality requires a systemic approach that considers the entire mineral composition rather than individual elements in isolation.

8.4 Bioavailability Is a Fundamental Principle of Nutritional Science

Modern nutrition recognizes that the mere presence of a nutrient does not guarantee its utilization by the human body.

Its absorption depends upon several factors, including:

- Chemical form;
- Solubility;
- Interactions with other minerals;
- Individual metabolism;
- Physiological conditions.

This concept, known as **bioavailability**, represents one of the cornerstones of contemporary nutritional science.

8.5 Water Quality Depends on Multiple Parameters

National and international organizations responsible for drinking water safety evaluate numerous parameters simultaneously, including:

- Microbiological quality;
- Absence of contaminants;
- Physicochemical composition;

- Element concentrations;
- pH;
- Total Dissolved Solids (TDS);
- Chemical stability;
- Compliance with drinking water regulations.

No single parameter, by itself, is sufficient to determine the overall quality of drinking water.

Only the combined evaluation of all relevant parameters provides a scientifically reliable assessment.

8.6 The Philosophy Behind ONE Technology

ONE Technology was developed with a clear and specific objective:

To preserve, within internationally recognized drinking water safety standards, the naturally occurring mineral richness of seawater while removing excess salts and potential contaminants, without compromising the complexity of the mineral system that nature has developed over billions of years.

This philosophy distinguishes ONE Technology from conventional desalination systems whose primary objective is simply to remove dissolved salts.

Our mission is to preserve a geochemical heritage created by nature itself, combining technological innovation, scientific responsibility, and rigorous quality control.

FINAL CONSIDERATIONS

Throughout human history, many of the greatest scientific advances have emerged when we learned to observe nature not as something to be replaced, but as a source of knowledge and inspiration.

The oceans represent one of the oldest, most complex, and most stable natural systems on Earth.

Over billions of years, geological, chemical, and biological processes have created a unique mineral balance capable of sustaining the largest and most diverse ecosystem on the planet.

Inspired by this natural intelligence, ONE Technology seeks to preserve that mineral richness through advanced desalination, purification, and quality control processes, producing drinking water with scientific responsibility while fully respecting internationally recognized safety standards.

We do not claim that water replaces a balanced diet, medical care, or healthy lifestyle choices.

Nor do we attribute therapeutic properties that have not been demonstrated through appropriate scientific research.

Our commitment is different.

It is to provide drinking water whose mineral composition respects the complexity developed by nature itself, produced through advanced technology, scientific transparency, and rigorous analytical control, enabling consumers, researchers, and partners to make informed decisions based upon reliable and scientifically grounded information.

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